```
# important nltk ( natural Language tool Kit ) libraries

import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]       date!
True
```

```
# word Tokenization

import nltk

nltk.download("punkt")
nltk.download("punkt_tab")

from nltk.tokenize import word_tokenize

text = "I really like this product because it is excellent."

words = word_tokenize(text)

print(words)
```

```
['I', 'really', 'like', 'this', 'product', 'because', 'it', 'is', 'excellent', '.']
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

```
'''
| Tag | Meaning         | Example  |
| --- | --------------- | -------- |
| NN  | Noun            | student  |
| NNS | Plural noun     | students |
| VB  | Verb            | play     |
| VBD | Past tense verb | played   |
| JJ  | Adjective       | good     |
| RB  | Adverb          | quickly  |
| PRP | Pronoun         | he       |
| DT  | Determiner      | the      |
 '''
```

```
'\n| Tag | Meaning         | Example  |\n| --- | --------------- | -------- |\n| NN  | Noun            | student  |\n| NNS
| Plural noun     | students |\n| VB  | Verb            | play     |\n| VBD | Past tense verb | played   |\n| JJ  | Adjecti
ve       | good     |\n| RB  | Adverb          | quickly  |\n| PRP | Pronoun         | he       |\n| DT  | Determiner
| the      |\n '
```

```
'''
| Word        | POS Tag | Meaning                                  |
| ----------- | ------- | ---------------------------------------- |
| **Ram**     | `NNP`   | Proper noun                              |
| **is**      | `VBZ`   | Verb – present tense, 3rd person singular |
| **a**       | `DT`    | Determiner                               |
| **good**    | `JJ`    | Adjective                                |
| **student** | `NN`    | Noun                                     |
| **and**     | `CC`    | Coordinating conjunction                 |
| **he**      | `PRP`   | Personal pronoun                         |
| **studies** | `VBZ`   | Verb – present tense, 3rd person singular |
| **Python**  | `NNP`   | Proper noun                              |
| **.**       | `.`     | Punctuation                              |
'''
```

```
'\n| Word        | POS Tag | Meaning                                    |\n| ----------- | ------- | ----------------------
------------------ |\n| **Ram**     | `NNP`   | Proper noun                                |\n| **is**      | `VBZ`   | Verb
— present tense, 3rd person singular |\n| **a**       | `DT`    | Determiner                                 |\n| **good**
```

```
# Parts of speech tagging

import nltk
from nltk.tokenize import word_tokenize

text = "Ram is a good student and he studies Python"

# Tokenization
words = word_tokenize(text)

# POS tagging
tags = nltk.pos_tag(words)

print("Word\t\tPOS Tag")
print("----------------------")

for word, tag in tags:
    print(word, "\t\t", tag)
```

```
Word            POS Tag
----------------------
Ram              NNP
is               VBZ
a                DT
good             JJ
student                  NN
and              CC
he               PRP
studies                  NNS
Python           NNP
```

```
# word Frequency - Basic Text analytics

from nltk.tokenize import word_tokenize
from nltk.probability import FreqDist

text = """
Python is easy to learn.
Python is powerful.
Python is useful for data science.
Data science is interesting.
"""

words = word_tokenize(text.lower())

# Keep only alphabetic words
words = [word for word in words if word.isalpha()]

frequency = FreqDist(words)

print(frequency.most_common())
```

```
[('is', 4), ('python', 3), ('data', 2), ('science', 2), ('easy', 1), ('to', 1), ('learn', 1), ('powerful', 1), ('useful', 1)
```